

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL (style)
ACADEX PRACTICE EXAMINATION
O-LEVEL · COMBINED SCIENCE

COMBINED SCIENCE 5006/2

PAPER 2 November 2019

2 hours

PAPER 2 — STRUCTURED QUESTIONS — ANSWER ALL — 2 HOURS

These are **original ACADEX questions** written to match ZIMSEC syllabus structure, mark allocations and command words. They are **not** leaked or copied official papers. This script is unique to November 2019 5006/2.

Additional materials: Answer all questions. The number of marks is given in brackets [].

INSTRUCTIONS: Write your answers in the spaces provided. You may use a calculator.

1. (a) What is meant by a biological catalyst? [2]<<
>>(b) Name the enzyme that digests starch and the product formed. [2]<<
>>(c) Describe the effect of increasing temperature on enzyme activity up to and beyond the optimum. [3]<<
>>(d) Explain the lock-and-key model of enzyme action. [3] **[10]**

2. (a) Write the balanced symbol equation for photosynthesis. [3]<<
>>(b) Name the green pigment and the organelle where it is found. [2]<<
>>(c) State two factors that can limit the rate of photosynthesis. [2]<<
>>(d) Describe a test for starch in a destarched leaf that has been exposed to light. [3] **[10]**

3. (a) Name the four chambers of the human heart. [2]<<
>>(b) Name the blood vessel that carries oxygenated blood from the lungs to the heart. [1]<<
>>(c) Describe the path of oxygenated blood from the lungs to a capillary in the body. [4]<<
>>(d) Give two differences between an artery and a vein. [3] **[10]**

4. (a) State where sperm are produced and where fertilisation occurs in humans. [2]<<
>>(b) Name two male secondary sexual characteristics. [2]<<
>>(c) Describe one barrier method of contraception and how it works. [3]<<
>>(d) Explain how the placenta is important to the fetus. [3] **[10]**

5. (a) Describe the arrangement and motion of particles in a solid and in a gas. [4]<<
>>(b) Name the change of state from solid to liquid and from gas to liquid. [2]<<
>>(c) Explain, in terms of particles, why a gas can be compressed but a liquid cannot. [4] **[10]**
6. (a) Name the two substances that must be present for iron to rust. [2]<<
>>(b) Give the chemical name of rust. [2]<<
>>(c) Describe two methods of preventing rusting and explain how each works. [4]<<
>>(d) Why is aluminium slower to corrode in air than iron? [2] **[10]**
7. (a) Define pressure and state its SI unit. [2]<<
>>(b) A force of 80 N acts on an area of 5 m². Calculate the pressure. [3]<<
>>(c) Explain why a sharp knife cuts more easily than a blunt knife for the same force. [3]<<
>>(d) State how liquid pressure changes with depth. [2] **[10]**
8. (a) State the rule for attraction and repulsion of magnetic poles. [2]<<
>>(b) Name two magnetic materials. [2]<<
>>(c) Describe how to plot the magnetic field of a bar magnet using a plotting compass. [3]<<
>>(d) Explain how an electromagnet can be made stronger. [3] **[10]**

END OF QUESTION PAPER

Worked solutions and mark-scheme notes are inside the ACADEX app (Extract & Study).